

Board-Free Multi-LiDAR Extrinsic Calibration for UAV-Anchored Roadside Sensing: Active Attitude Excitation as a Necessary Condition

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Abstract—Roadside and V2X deployments increasingly rely on multiple LiDARs with little or no field-of-view overlap, anchored by UAV RTK trajectories rather than calibration targets. We show that board-free body-centric calibration is governed not only by sensor geometry but by *flight attitude diversity*: the empirical body centroid forms an aspect-dependent bias vector field on the relative bearing sphere u_B . Under concentrated aspect this field degenerates with extrinsic translation and silently pollutes estimates (~ 211 mm in simulation); active 3D attitude excitation (POI tidal-lock flight) breaks the degeneracy and restores decimeter-level accuracy (~ 20 mm in simulation). We present a separable two-stage estimator with per-LiDAR independent Stage-2 solves, an aspect gate with centroid fallback, and a relative-extrinsic covariance analysis under coherent RTK bias ($0.166\times$ in simulation). Field validation on diagonal dual-Ruby intersection data is structured but pending.

Index Terms—LiDAR calibration, extrinsic calibration, UAV, RTK, board-free calibration, multi-sensor systems

I. INTRODUCTION

Intersection-scale cooperative perception requires multiple roadside LiDARs whose fields of view barely overlap, if at all. Classical target-based or hand-eye pipelines assume shared views, calibration boards, or dedicated calibration yards—assumptions that fail at urban corners where sensors sit on diagonal posts tens of meters apart. UAV-RTK trajectories offer a natural anchor: a single flying platform carries the body frame through the combined visibility of all sensors, enabling *board-free* calibration from vehicle point clouds alone.

The counter-intuitive finding of this work is that board-free accuracy is dominated not by LiDAR range noise or RTK centimeter fixes, but by **how diversely the UAV attitude is excited during the calibration flight**. The empirical body centroid is not a fixed offset: it is an *aspect-dependent bias vector field* on the relative bearing u_B . When aspect is concentrated, this field aligns with extrinsic translation and *silently* corrupts calibration—joint optimization of the lever arm cannot remove it. Only active 3D attitude excitation (sector POI tidal-lock plus pitch/yaw diversity) decorrelates the field so that its world-frame mean can be estimated and removed, recovering sub-decimeter extrinsics in simulation.

Memory line: *board-free calibration accuracy is determined by flight attitude diversity, not by the sensor.*

A. Contributions

We contribute:

- 1) A **separable two-stage framework**: Stage-1 fits a continuous-time RTK-anchored trajectory $T_{WB}(t)$;

Stage-2 freezes the trajectory and calibrates each LiDAR extrinsic independently—enabling diagonal multi-LiDAR deployment with zero FOV overlap.

- 2) A **mechanistic account of board-free identifiability**: the u_B bias vector field, a 2×2 degeneracy table (aspect $\times$ attitude), and the conclusion that **active 3D attitude excitation is a necessary condition** for unbiased board-free calibration (simulation).
- 3) A **relative-extrinsic covariance analysis** (§7.2): under a coherent systematic RTK bias assumption, $\text{Cov}(T_{\text{rel}})$ scales as $0.166\times$ absolute covariance; white per-epoch noise composes at $3.101\times$ (simulation), with explicit scope boundaries for field transfer.
- 4) **Prais–Winsten RTK whitening** for high-rate Stage-1 trajectory fitting, avoiding uniform covariance inflation that destroys low-frequency observability.
- 5) **Field validation** on diagonal dual-Ruby intersection data: [TODO: field campaign A–D: board-free vs sphere-anchor, coherent-bias residual, POI flight feasibility — see Sec. VIII].

Sphere targets (scope boundary). Throughout this paper, retroreflective sphere targets appear *only* as an independent ground-truth ruler to score board-free error in field experiments. They are **not** a fallback calibration method; board-free remains the sole calibration pipeline.

II. RELATED WORK

A. Continuous-time and spline-based calibration

Continuous-time trajectory representations coupled with batch factor-graph optimization are standard for multi-sensor extrinsic calibration [1], [2]. Our Stage-1 trajectory fit follows this lineage but *separates* trajectory estimation from per-sensor Stage-2 extrinsic refinement, avoiding cross-sensor Ceres block coupling that can leak state between LiDAR solves.

B. Roadside, V2X, and multi-LiDAR systems

Infrastructure sensing at intersections motivates long-baseline, low-overlap multi-LiDAR layouts [3]. Prior work often assumes overlapping views or shared calibration targets. We address the diagonal-post regime (~ 70 m baseline, negligible FOV overlap) with UAV-RTK anchoring and per-sensor separable solves.

C. Target-based vs. targetless calibration

Target-based LiDAR–camera and LiDAR–IMU calibration remains the industrial default [4]. Targetless methods exploit scene structure or motion constraints; body-centric board-free calibration uses the vehicle hull as a primitive without a physical board. **Distinction:** we do not propose sphere targets as an alternative pipeline; spheres are reserved for field ground-truth evaluation only (Sec. VIII).

D. UAV-assisted and cooperative calibration

UAV platforms with RTK positioning enable mobile anchoring of roadside sensors [5]. Related trajectory-matching approaches align sensor frames via shared motion [6]. Our contribution is orthogonal: even with a perfect RTK trajectory, board-free calibration fails without sufficient attitude excitation—a *flight-design* requirement not captured by trajectory matching alone.

E. Relation to prior work on degeneracy

Anonymous prior work on coplanar multi-LiDAR setups identifies *geometric rank deficiency* in joint Fisher information when sensors and the body path are nearly coplanar (simulation $\lambda_{\min} \approx 0$ vs. multi-layer $\lambda_{\min} = 1.074$). **This paper addresses a different degeneracy:** aspect-concentrated body-centroid bias aligns with extrinsic translation (~ 211 mm silent pollution in simulation), which joint lever-arm optimization cannot remove. The two failure modes are complementary: coplanar geometry vs. aspect–attitude coupling.

III. SEPARABLE TWO-STAGE CALIBRATION FRAMEWORK

A. Problem setup

Consider L LiDARs rigidly mounted on a UAV body frame $\{B\}$. Each LiDAR ℓ has unknown extrinsic $T_{L_\ell B} \in SE(3)$. During a calibration flight, RTK fixes the world pose trajectory $T_{WB}(t)$ and each LiDAR observes body-surface points. We seek $\{T_{L_\ell B}\}$ and a consistent $T_{WB}(t)$ without calibration boards.

B. Stage-1: RTK-anchored trajectory fitting

Stage-1 estimates a continuous-time body trajectory $\hat{T}_{WB}(t)$ from RTK position measurements (and optional attitude streams), using Prais–Winsten differenced whitening at high rate (Sec. VI). The output trajectory is **frozen** before any extrinsic refinement.

C. Stage-2: frozen-trajectory extrinsic refinement

Given frozen $\hat{T}_{WB}(t)$, Stage-2 minimizes a Ceres cost over $T_{L_\ell B}$ using body-cluster centroid factors and, when the aspect gate opens, an observed-mean lever $\mathbf{p}_B$ (Sec. IV). Each LiDAR ℓ is solved in an **independent** `ExtrinsicRefiner` scope—no shared `CalibrationEstimator` state across sensors (`multi_lidar_body_calibration`, `commit 3fb7ddf`).

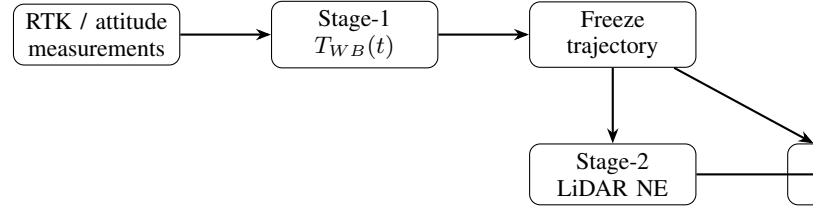


Fig. 1. Separable two-stage pipeline: shared Stage-1 trajectory, independent per-LiDAR Stage-2 Ceres scopes, post-hoc relative composition. [TODO: replace with publication-quality figure from figure_table_plan.md Fig. 1].

D. Multi-LiDAR composition without FOV overlap

For diagonal deployment (e.g., NE and SW corner posts), sensors need not see the same scene. Per-sensor absolute extrinsics $\hat{T}_{L_\ell W}$ are estimated separately; the relative transform is composed post hoc:

$$\hat{T}_{L_{SW} \leftarrow L_{NE}} = \hat{T}_{L_{SW} W} \hat{T}_{L_{NE} W}^{-1}. \quad (1)$$

Regression tests confirm production multi-LiDAR NE/SW errors match single-sensor in-process solves (12.149 mm / 42.041 mm, simulation).

E. Framework diagram

F. Production modules

Key implementations in `clic_calib`: `Stage1TrajectoryFitter` (PW-whitened RTK trajectory), `body_gated_calibration` (gated observed-mean $\times 3$ + guards), `observed_mean_gate` (`yaw_cv` ≥ 0.65 + POI path), `multi_lidar_body_calibration` (per-sensor Stage-2 loop), and `relative_extrinsic` (§7.2 covariance and center-reg metric).

IV. BOARD-FREE MECHANISM: ASPECT-DEPENDENT BIAS ON u_B

This section is the oral core: **why** board-free calibration succeeds or fails.

A. Bias as a vector field on relative bearing

At each body-cluster epoch i , define the empirical centroid bias

$$\mathbf{b}_{B,i} = \hat{\mathbf{c}}_{B,i}^{\text{emp}} - \mathbf{L}_{B \rightarrow \text{body}}^{\text{nom}}, \quad (2)$$

where $\hat{\mathbf{c}}^{\text{emp}}$ is the centroid of *visible* sampled points (not the nominal box center plus noise). The relative bearing u_B lives on the unit sphere; $\mathbf{b}_B$ is a **vector field** over flown aspects, not a constant offset.

The observed-mean lever $\mathbf{p}_B$ estimates the **mean of this field** over the aspect distribution actually flown:

$$\mathbf{p}_B \approx \mathbb{E}_{u_B} [\mathbf{b}_B(u_B)]. \quad (3)$$

B. Diagnostic: Flight-D correlation structure (simulation)

Wide u_B sweep with decoupled orbit attitude (seed 13025): Opposite signs on x vs. y indicate near-sinusoidal coupling to u_B azimuth; range-only bias models are rejected.

TABLE I
ASPECT-BIAS COUPLING ON FLIGHT-D (SIMULATION).

Sensor	u_B az std [°]	varying RMS [mm]	$r(u_B, \text{bias}_x)$	$r(u_B, \text{bias}_y)$
NE	14.1	110	+0.53	-0.68
SW	17.6	124	-0.64	+0.69
P1.5 ref	—	22	—	—

TABLE III
CONCENTRATED VS. DIVERSE ASPECT (SIMULATION).

$\ \text{trans}\ $ [mm]	Gate	Note
211.4	—	bias $\parallel t_{LW}$
63.5	—	partial yaw help
18.5	ON	$\mathbf{p}_B$ decorrelates
~211	—	no-op
20.4	ON	target

TABLE II
DEGENERACY TABLE: u_B SWEEP VS. POI-LOCK $\times$ R_{WB} DIVERSITY (SIMULATION).

	R_{WB} sweeps	R_{WB} near-locked
u_B sweeps (Flight-D)	NE obs 60.3 / SW 89.9 mm; rel 311 mm	—
u_B POI-locked (Flight-W)	NE 12.2 / SW 42.0 mm [†] ; rel rot 0.12°	Flight-J gate OFF: 31.5 mm

[†]SW 42 mm = sector geometric anisotropy, not estimator leak (Sec. VII).

TABLE IV
WORLD-FRAME MEAN BIAS NORM VS. FLIGHT REGIME (SIMULATION).

Regime	$\ \text{mean}(\mathbf{v})\ _h$ [mm]	pitch std [°]	yaw_cv
Concentrated aspect	~220	~0	~0
Diverse (P1.5)	~51	~16	~0.78

C. 2×2 degeneracy: aspect $\times$ attitude

Board-free $\mathbf{p}_B$ identifiability closes on a 2×2 table:

1) *Concentrated aspect: silent pollution:* Phase 1.5 ablation (simulation):

Joint $\mathbf{p}_B$ no-op: under concentrated aspect, joint optimization cannot uncouple bias from t_{LW} ; only aspect diversity plus the gate unlocks observed-mean.

2) *Gate as mechanism: Flight-Y (simulation):* When yaw_cv = 0.32 (below 0.65 threshold) and pitch is low, the aspect gate **closes**; observed-mean is not applied and the system falls back to centroid-only at 165.0 mm—not a 1243 mm blow-up. The gate embodies *self-awareness*: insufficient excitation $\Rightarrow$ do not pretend $\mathbf{p}_B$ is identifiable.

D. Frame-wise quantitative core (simulation)

Per frame, the world-frame bias driver is

$$\mathbf{v}_i = R_{LW} R_{WB}(t_i) \mathbf{b}_{\text{const}}. \quad (4)$$

- **Horizontal 220→51 mm:** yaw sweeps average horizontal components of $R_{WB}\mathbf{b}$ —diversity in *attitude*, not translation alone.
- **Vertical ~19.5 mm residual:** body-Z component of $\mathbf{b}_{\text{const}}$; yaw cannot average it; observed-mean $\mathbf{p}_B$ removes it.

Tidal lock audit (simulation): P1.5 NearFieldHighAspect yaw-orbit std 0.0° (locked); Flight-D sector-0: 110.4° (not locked).

E. Necessary condition (conclusion sentence)

Active 3D attitude excitation (diverse pitch/yaw plus POI sector lock where needed) is a necessary condition for unbiased board-free extrinsic calibration—not a refinement.

F. Mechanism figure (placeholder)

G. Engineering guardrails

Production applies observed-mean only if Ceres cost $\leq$ centroid cost **and** $\|\text{trans}\|_{\text{obs}} \leq \|\text{trans}\|_{\text{cent}}$ (simulation ground-truth guard in phase15_ablation_common.hpp). Aspect gate: yaw_cv $\geq$ 0.65 (+ POI path) $\Rightarrow$ try observed-mean $\times 3$; P1.5 opens at yaw_cv $\approx$ 0.78.

V. RELATIVE EXTRINSIC UNCERTAINTY (§7.2)

Absolute extrinsics $\hat{T}_{LW}$ inherit RTK position uncertainty. For long-baseline diagonal layouts, the **relative** transform $\hat{T}_{LW \leftarrow LNE}$ is often the quantity of interest for fusion. We analyze how RTK error modes propagate through independent Stage-2 solves (simulation, sealed @ commit 33f9647).

A. Composition and metrics

Relative transform is composed post hoc (Sec. III). We report:

- **center-reg (obs)** — headline metric, rotation-dominated, no 70.7 m lever amplification (40.9 mm @ rep, simulation);
- rel-trans (obs) — deprecated lever metric (49.0 mm, simulation).

B. RTK perturbation dichotomy (simulation)

Monte Carlo ($N=20$, Flight-W 0.5 Hz, independent Stage-2 per LiDAR):

Claims (simulation, scoped):

- Under a **coherent systematic-bias assumption**, relative covariance is tighter than absolute (0.166 $\times$).
- White RTK noise **composes** in the relative frame (3.101 $\times$)—not an architecture defect.

C. Temporal overlap at sector handoff (simulation)

Brief temporal overlap can tighten relative UQ without spatial FOV overlap.

[TODO: Fig. 2 mechanism panel: (a) u_B sphere vector field + Flight-D scatter $r \approx \pm 0.5\text{--}0.7$; (b) 2×2 degeneracy table; (c) $\|\text{mean}(\mathbf{v})\|$ curve 220→51 mm + 19.5 mm vertical bar; see figure_table_plan.md]

Fig. 2. Board-free mechanism at a glance (placeholder layout). Panel (a): aspect-dependent bias on u_B . Panel (b): degeneracy table. Panel (c): frame-wise world-frame averaging under attitude diversity.

TABLE V
RTK INJECTION CAUSAL CHAIN (SIMULATION).

Injection	$\text{tr Cov}(\delta t_{\text{rel}})/\text{mean Cov}(\delta t_{\text{abs}})$	$r(\ \delta p\ _{\text{NE}}, \ \delta p\ _{\text{SW}})$	Path	Mechanism	10 Hz antenna RMSE [mm]
White noise (per epoch)	3.101 $\times$	0.40	v2 uniform RTD scaling	uniform RTD scaling	36.0
Coherent system bias	0.166 $\times$	1.00	PW differenced whitening	PW differenced whitening	27.3
			0.5 Hz (PW auto-off, $\Delta t \geq 0.2$ s)		23.7

TABLE VI
COHERENT-BIAS REL/ABS RATIO VS. OVERLAP (SIMULATION).

Geometry	coherent rel/mean(abs)
Serial Flight-W	0.166 $\times$
+10 s overlap at handoff	0.15 $\times$

TABLE VII
STAGE-1 RTK PATHS (SIMULATION).

TABLE VIII
P1.5 SINGLE-LIDAR BOARD-FREE (SIMULATION).

Metric	Value [mm]
Centroid-only $\ \text{trans}\ $	63.5
Observed-mean $\times 3$	18.5
Joint-opt reference	20.4
GT varying RMS (bias field)	~ 22

D. Scope boundary (mandatory)

The 0.166 $\times$ ratio depends on injected **rigid global bias** (RtkMcInjectionMode::kCoherentSystemBiasOnly, supports trajectory quality for Stage-2 but does not replace the attitude-excitation requirement of Sec. IV. $\sigma_h=50$ mm, $\sigma_v=100$ mm, one draw per epoch block). Real RTK includes drift, colored noise, and lever-arm coupling—**not** perfect common-mode. **Do not cite 0.166 $\times$ as a field guarantee.** Field cancellation is expected to be **weaker**; hardware relative-extrinsic residual under coherent bias is the empirical test (Sec. VIII, Table XII).

VI. PRAIS–WINSTEN RTK WHITENING (STAGE-1)

High-rate RTK (10 Hz) exhibits serial correlation in position residuals. Stage-1 must whiten without destroying low-frequency observability.

A. Why uniform Σ inflation fails

Uniform covariance scaling (wrong AR(1) shortcut) inflates all frequencies and yields ~ 125 mm RMSE—unacceptable (simulation audit, rejected).

B. PW differenced whitening

We apply Prais–Winsten (PW) differenced whitening: estimate lag-1 correlation ρ from RTK residuals and decorrelate successive epochs. Results (simulation, Stage-1 antenna RMSE):

PW reduces 36.0→27.3 mm at 10 Hz; at 0.5 Hz independent epochs PW disables automatically ($\Delta t \geq 0.2$ s). This section

VII. SIMULATION VALIDATION

All numbers in this section are (simulation) unless noted. Reproduce: `scripts/verify_frozen_baseline_isolation.sh @ commits f03c399 / 3fb7ddf / 7f5c27c; seed 13025; noise config/noise_model.yaml.`

A. Single-LiDAR Phase 1.5 (NearFieldHighAspect)

Diverse aspect ($\text{yaw}_{\text{cv}} \approx 0.78$, $\text{pitch}_{\text{std}} \approx 16^\circ$) enables $\mathbf{p}_B$; centroid floor ~ 64 mm, observed-mean $\sim 18\text{--}20$ mm.

B. Diagonal dual-LiDAR @ Flight-W

Geometry: NE/SW posts at $(\pm 25, \pm 25)$ m, baseline 70.7 m, POI serial lock $t \in [0, 45) \rightarrow \text{NE}$, $t \in [45, 90) \rightarrow \text{SW}$.

1) *Honest NE/SW asymmetry*: SW ~ 42 mm is **confirmed sector geometric anisotropy**, not estimator leak: per-sensor independent `ExtrinsicRefiner` scopes reproduce frozen values. Mechanism: SW sightline vs. body main tilt axis $\Rightarrow$ larger lever projection of $\mathbf{b}_{\text{const}}$. **Paper obligation**: present both sectors; do not claim symmetric accuracy.

TABLE IX
DUAL-LiDAR @ 0.5 Hz (SIMULATION).

	NE obs [mm]	SW obs [mm]	rel rot [°]	center-reg [mm]
Frozen gate	12.2	42.0	0.12	NE 41
Production @ 3fb7ddf	12.149	42.041	—	SW—

TABLE X
COPLANAR ABLATION VS. MULTI-LAYER (SIMULATION).

Configuration	$\lambda_{\min}$
Multi-layer (well observed)	1.07357 (≈ 1.074)
Coplanar (degenerate)	≈ 0 (numerical null)

TABLE XI
FIELD BOARD-FREE ACCURACY VS. SPHERE-ANCHOR GROUND TRUTH.

	Δ_{trans} [mm]	Δ_{rot} [°]
NE 41	[TODO: field campaign A: NE Δ_{trans} vs sphere-anchor]	[TODO: field campaign A: NE Δ_{rot} vs sphere-anchor]
SW—	[TODO: field campaign A: SW Δ_{trans} vs sphere-anchor]	[TODO: field campaign A: SW Δ_{rot} vs sphere-anchor]

TABLE XII
FIELD TEST OF §7.2 COHERENT-BIAS CANCELLATION (VS. SIMULATION 0.166×).

Metric	Simulation	
tr Cov(δt_{rel})/mean Cov(δt_{abs})	0.166×	[TODO: field campaign D: m...]
$r(\ \delta p\ _{\text{NE}}, \ \delta p\ _{\text{SW}})$	1.00 (coherent)	[TODO: field campaign D: m...]
center-reg under coherent perturbation	—	[TODO: field campaign D: m...]

C. Observability gate (joint FIM $\lambda_{\min}$)

Distinct from aspect degeneracy (Sec. IV): coplanar geometry yields rank deficiency in joint FIM; multi-layer flight restores observability.

D. Two-stage pipeline sanity (Config C)

Centimeter-level split 50/50; mean $|LW|$ translation 6.556 mm (simulation).

E. Simulation sign-off status

Simulation phase **SEALED** after 33f9647 + 5f19793. No further synthetic tuning; field data per blueprint doc/multi_lidar_board_free_blueprint.md.

VIII. FIELD EXPERIMENTS

Status: structured placeholders only — no field numbers in this draft.

This section defines what we will measure on real intersection data. Sphere targets are used **only** as an independent ground-truth ruler to score board-free error; they are **not** a calibration fallback.

A. Experimental setup (placeholder)

- **Site:** urban intersection, diagonal dual Ruby LiDAR posts, baseline ~ 70 m, heights 3–6 m, negligible FOV overlap. [TODO: field campaign A: actual site name, GPS coordinates, post survey]
- **Platform:** DJI M3E + RTK + PSDK LiDAR @ 10 Hz. [TODO: field campaign A: firmware versions, antenna lever-arm survey]
- **Flight:** POI tidal-lock sectors per simulation Flight-W; attitude diversity per P1.5 recipe. [TODO: field campaign B: flight logs, POI sector timing, weather]
- **Ground truth:** retroreflective sphere anchors surveyed by total station (independent of board-free pipeline). [TODO: field campaign C: sphere positions, survey uncertainty]

B. Sphere-anchor evaluation protocol (method, not calibration)

On the **same** flight segment:

- 1) Survey sphere centers (total station) $\Rightarrow$ ground-truth extrinsic ruler.
- 2) Run board-free calibration once (sole method).
- 3) Report Δ_{trans} , Δ_{rot} , center-reg vs. sphere-derived truth.

Spheres measure error; they do not replace board-free.

C. Table XI: board-free vs. sphere truth

D. Table XII: coherent-bias residual (total-station check)

E. Research questions for field (results pending)

- 1) **Real point-cloud bias magnitude:** does empirical $\mathbf{b}_B(u_B)$ on live data match simulation field shape? [TODO: field campaign B: per-epoch bias vs u_B scatter plot data]
- 2) **Board-free field accuracy:** sub-decimeter NE and honest SW sector asymmetry at intersection scale? [TODO: field campaign A: Table XI cells]
- 3) **Coherent-bias residual:** is field rel/abs ratio weaker than 0.166×? [TODO: field campaign D: Table XII cells]
- 4) **POI flight feasibility:** can operators execute tidal-lock sectors without degrading RTK or safety margins? [TODO: field campaign B: POI execution notes + gate yaw_cv statistics]

F. Field figures (placeholder)

IX. CONCLUSION

We presented a separable RTK-anchored framework for board-free multi-LiDAR extrinsic calibration at intersections with negligible FOV overlap. The central mechanism is an aspect-dependent body-centroid bias vector field on u_B : under concentrated aspect it degenerates with translation and silently pollutes estimates (~ 211 mm in simulation); active 3D attitude excitation breaks the degeneracy and restores ~ 20 mm accuracy (simulation). An aspect gate provides engineering self-awareness when excitation is insufficient. Relative extrinsic uncertainty under coherent RTK bias can be tighter than

[TODO: field campaign A: intersection layout +
diagonal Ruby posts + flight path]

Fig. 3. Field deployment schematic (placeholder).

[TODO: field campaign B: real u_B -bias scatter vs
simulation Fig. 2a]

Fig. 4. Field aspect-bias diagnostic (placeholder).

absolute ($0.166\times$, simulation, scoped assumption); white noise composes at $3.101\times$.

Field validation on diagonal dual-Ruby data is structured in Sec. VIII and remains to be filled.

A. Future work

- **Field campaigns** A–D: Tables XI–XII.
- **Road-network scale:** corridor–intersection mixed layouts, city-scale multi-LiDAR graphs (journal extension).
- **Unified target framework:** a future T-ITS line may treat sphere/board targets as a *calibration method* for depot-style setups—orthogonal to this paper, where spheres are evaluation rulers only.
- **Online FIM / adaptive planning:** reuse of `CalibrationEstimator` across missions must address deferred `ClearProblem` leak (known issue @ 3fb7ddf).

ACKNOWLEDGMENT

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